Exercise 2: Worked solutions

Mathematics for Engineers I · Module 62765 · Exercise Sheet 2

From Fundamentals to AI Apocalypse · personal study guide

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Exercise 2.1: Absolute values and inequalities

Absolute value is distance from zero. Splitting at sign changes turns these expressions into ordinary inequalities on intervals.

Connected topic: Absolute values and inequalities (§2.3) explain the distance interpretation and safe case splitting.

1a(i): Sketch $x^2 - 2|y| > 1$

Try first: isolate $|y|$, then check where the resulting upper bound is non-negative.

Answers: see "Exercise 2.1: Absolute values and inequalities" under Answers at the end.

1a(ii): Sketch $|x| + |y| = 2$

Try first: split the plane into quadrants and find where the boundary meets each axis.

Answers: see "Exercise 2.1: Absolute values and inequalities" under Answers at the end.

1b(i): Solve $|x + 1| - |x - 1| = 1$

Try first: split at -1 and 1, where the expressions inside the absolute values change sign.

Answers: see "Exercise 2.1: Absolute values and inequalities" under Answers at the end.

1b(ii): Solve $\sqrt{x - 12} < x - 2$

Try first: write the square-root domain and make sure both sides are non-negative before squaring.

Answers: see "Exercise 2.1: Absolute values and inequalities" under Answers at the end.

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Exercise 2.2: Supremum and infimum

A supremum is a least upper bound; it need not be a member of the set. A maximum must belong to the set, and the same distinction applies to infimum and minimum.

Connected topic: Real numbers, completeness, supremum and infimum (§2.2) give the bound definitions and the ε-characterisation used here.

2a(i): Find the bounds and extrema of $M_1 = (0, 1]$

Try first: identify the numbers approached at both ends and check which endpoints the interval includes.

Answers: see "Exercise 2.2: Supremum and infimum" under Answers at the end.

2a(ii): Find the bounds and extrema of $M_2 = \{\frac{1}{n} : n \in \mathbb{N}\}$

Try first: list the first few values and determine their limit as n grows.

Answers: see "Exercise 2.2: Supremum and infimum" under Answers at the end.

2a(iii): Find the bounds and extrema of $M_3 = \{ (-1)^n(1 + \frac{1}{n}) : n \in \mathbb{N} \}$

Try first: separate even n from odd n; their signs differ, so each subsequence controls a different extreme.

Answers: see "Exercise 2.2: Supremum and infimum" under Answers at the end.

2b: Show that $M = \{x \in \mathbb{Q} : x^2 < 2\}$ has infinitely many rational upper bounds but no rational supremum

Try first: show that every rational greater than $\sqrt{2}$ is an upper bound, then ask whether such a bound can be the least rational one.

Answers: see "Exercise 2.2: Supremum and infimum" under Answers at the end.

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Exercise 2.3: Sums, products, binomial coefficients

Use index changes to expose patterns, cancellation to simplify sums, and combinations to count pairs or selections.

Connected topic: Sums, products and binomial coefficients (§2.4) define the notation, geometric sums and counting rules used below.

3a: Write the odd square numbers from 121 to 2401 as a sum

Try first: represent the odd bases as a sequence with step 2, starting at 11 and ending at 49.

Answers: see "Exercise 2.3: Sums, products, binomial coefficients" under Answers at the end.

3b(i): Simplify the difference of two sums

Try first: shift the first index so both sums contain the same powers of x, then compare coefficients.

Answers: see "Exercise 2.3: Sums, products, binomial coefficients" under Answers at the end.

3b(ii): Compute the nested sum and product

Try first: evaluate the inner product for each m; look for a zero factor before multiplying numbers.

Answers: see "Exercise 2.3: Sums, products, binomial coefficients" under Answers at the end.

3b(iii): Simplify the sum of C(n, k) over k

Try first: use the binomial theorem with both entries equal to 1.

Answers: see "Exercise 2.3: Sums, products, binomial coefficients" under Answers at the end.

3c: Find the number of guests from 66 clinks

Try first: one clink corresponds to one unordered pair of distinct guests.

Answers: see "Exercise 2.3: Sums, products, binomial coefficients" under Answers at the end.

3d: Birthday probability for 27 students

Try first: calculate the complement event that every birthday is different, then subtract from 1.

Answers: see "Exercise 2.3: Sums, products, binomial coefficients" under Answers at the end.

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Exercise 2.4: Order axioms

Each result follows from the field and order axioms, not from assuming familiar arithmetic laws. Use uniqueness of inverses, distributivity, trichotomy, and compatibility of order with addition and positive multiplication.

Connected topic: Real numbers as an ordered field (§2.2) lists the axioms from which each identity and inequality is derived.

4i: Prove $a \cdot 0 = 0$

Try first: write 0 as $0 + 0$ and use distributivity, then cancel a term from both sides.

Answers: see "Exercise 2.4: Order axioms" under Answers at the end.

4ii: Prove $-a = (-1) \cdot a$

Try first: show that $(-1) \cdot a$ added to a equals zero, then use uniqueness of the additive inverse.

Answers: see "Exercise 2.4: Order axioms" under Answers at the end.

4iii: Prove $(-a) \cdot (-b) = a \cdot b$

Try first: replace each negative by multiplication by -1; establish $(-1)(-1) = 1$ from part ii).

Answers: see "Exercise 2.4: Order axioms" under Answers at the end.

4iv: Prove $ab = 0 \Rightarrow a = 0$ or $b = 0$

Try first: assume $a \neq 0$ so its multiplicative inverse exists, then multiply the equation by that inverse.

Answers: see "Exercise 2.4: Order axioms" under Answers at the end.

4v: Prove $1 > 0$

Try first: use trichotomy and rule out $1 = 0$ and $1 < 0$.

Answers: see "Exercise 2.4: Order axioms" under Answers at the end.

4vi: Prove $a^2 > 0$ for $a \neq 0$

Try first: split into $a > 0$ and $a < 0$, and express the second square using $-a$.

Answers: see "Exercise 2.4: Order axioms" under Answers at the end.

4vii: Show $a < b$ and $c < 0$ imply $ac > bc$

Try first: translate $a < b$ into a positive difference, and identify the sign of $c(b - a)$.

Answers: see "Exercise 2.4: Order axioms" under Answers at the end.

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Self-Study 2.5: Absolute values and inequalities

The first condition describes a diamond in the plane; the one-variable inequalities are handled by splitting intervals and checking domains before squaring.

Connected topic: Absolute values and inequalities (§2.3) connect distance regions with interval solutions.

5a: Sketch $|x| + |y - 2| < 5$

Try first: interpret each term as a distance from a coordinate, then locate the four axis-parallel intercepts.

Answers: see "Self-Study 2.5: Absolute values and inequalities" under Answers at the end.

5b(i): Solve $|x + 3| + |x - 3| > 8$

Try first: inside $[-3,3]$ the two distances add to a constant; outside it they grow with $|x|$.

Answers: see "Self-Study 2.5: Absolute values and inequalities" under Answers at the end.

5b(ii): Solve $\sqrt{(x - 4)(x - 1)} \leq 2 - x$

Try first: require a non-negative radicand and a non-negative right side before squaring.

Answers: see "Self-Study 2.5: Absolute values and inequalities" under Answers at the end.

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Self-Study 2.6: Sets, products and complements

Use ordered-pair membership for products and remember that every complement depends on its declared universe.

Connected topic: Set inclusion, Cartesian products and complements (§1.2) define the membership tests used below.

6a: Prove $A \times B \subset C \times D$ when $A \subset C$ and $B \subset D$

Try first: take an arbitrary ordered pair from $A \times B$ and track each coordinate separately.

Answers: see "Self-Study 2.6: Sets, products and complements" under Answers at the end.

6b: Is the converse true?

Try first: ask what happens when one Cartesian-product factor is empty, so the product contains no pairs.

Answers: see "Self-Study 2.6: Sets, products and complements" under Answers at the end.

6c: Compute the complements and difference for $A = [1,3)$, $B = (2,5]$ in $X = \mathbb{R}$

Try first: mark each endpoint, then form the union and intersection before taking their complements relative to $\mathbb{R}$.

Answers: see "Self-Study 2.6: Sets, products and complements" under Answers at the end.

6d: Find the complement of $[1,3] \times [1,2]$ in $\mathbb{R}^2$

Try first: a point is outside the rectangle if at least one coordinate lies outside its own closed interval.

Answers: see "Self-Study 2.6: Sets, products and complements" under Answers at the end.

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Self-Study 2.7: Sums, products, binomial coefficients

Match each term to an index, telescope products before expanding them, and use combinations when order does not matter.

Connected topic: Sums, products and binomial coefficients (§2.4) supply the geometric, telescoping and counting identities.

7a: Write $10^2 - \frac{1}{9^2} + 8^2 - \frac{1}{7^2} + \ldots + 2^2 - \frac{1}{1^2}$ as a sum

Try first: choose an index that makes the positive bases descend by 2 from 10 and the denominators descend from 9.

Answers: see "Self-Study 2.7: Sums, products, binomial coefficients" under Answers at the end.

7b(i): Simplify the difference of geometric sums

Try first: write out the powers in both sums and look for the unmatched term.

Answers: see "Self-Study 2.7: Sums, products, binomial coefficients" under Answers at the end.

7b(ii): Simplify the product

Try first: factor the numerator $j(j+1) - 2$ and separate the product into two telescoping ratios.

Answers: see "Self-Study 2.7: Sums, products, binomial coefficients" under Answers at the end.

7b(iii): Simplify the alternating binomial sum

Try first: expand $(1 + x)^n$ and substitute $x = -1$.

Answers: see "Self-Study 2.7: Sums, products, binomial coefficients" under Answers at the end.

7c: Pick six marbles with at most two striped marbles

Try first: remember there are only four white marbles, so any six-marble selection must include at least two striped ones.

Answers: see "Self-Study 2.7: Sums, products, binomial coefficients" under Answers at the end.

7d: Find the members from 272 spoken riddles

Try first: distinguish a greeting spoken from one unordered pair. Each member speaks once to each of the other members.

Answers: see "Self-Study 2.7: Sums, products, binomial coefficients" under Answers at the end.

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Answers

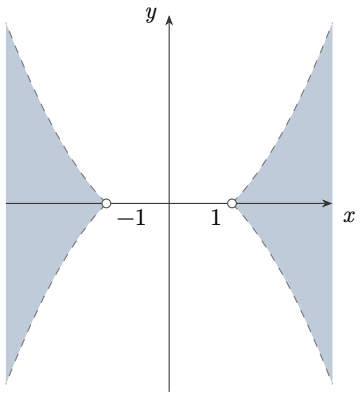
Exercise 2.1: Absolute values and inequalities

1a(i) · Two unbounded curved bands

1. Move the absolute-value term to the right: $2|y| < x^2 - 1$. Since the left side is non-negative, solutions require $x^2 - 1 > 0$, so $|x| > 1$.
2. For those x, divide by 2 and remove the absolute value as a two-sided bound:

$$-\frac{x^2 - 1}{2} < y < \frac{x^2 - 1}{2}, \text{ for } x < -1 \text{ or } x > 1$$

3. The solution is the interior between the curves $y = \pm\dfrac{x^2 - 1}{2}$ on the two outer sides $|x| > 1$. The curves and the vertical cutoffs $x = \pm 1$ are excluded. At $|x| = 1$ the only possible y would be 0, but the original inequality is strict and fails there.
4. **Check:** at $x = 2$, the condition is $|y| < \dfrac{3}{2}$; at $x = 0$ it has no solution.



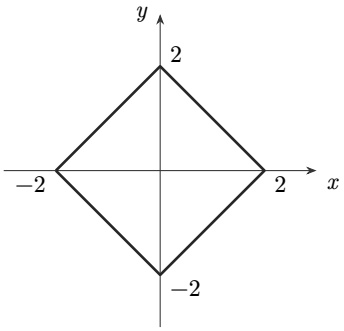
Boundary of $x^2 - 2|y| > 1$. The shaded bands are the solution set; the dashed curves and the open points at $(\pm 1, 0)$ are excluded, and the bands lie between the curves where $|x| > 1$. The gap for $-1 < x < 1$ has no solutions.

Made

Exercise 2.1: Absolute values and inequalities

1a(ii) · A diamond boundary

1. On each quadrant, replace absolute values by the corresponding signed coordinates. For example, when $x \geq 0$ and $y \geq 0$, the equation is $x + y = 2$.
2. Setting one coordinate to zero gives the four vertices $(2, 0)$, $(0, 2)$, $(-2, 0)$, and $(0, -2)$. Between consecutive vertices the equation is linear.
3. Join those points with straight segments and include the whole boundary, because equality is required. The enclosed diamond itself is not the requested set.
4. **Check:** $(1, 1)$ lies on it; $(0, 0)$ and $(2, 2)$ do not.



$|x| + |y| = 2$. The four straight sides join the intercepts. The equality keeps only this outline, not the region inside it.

Exercise 2.1: Absolute values and inequalities

1b(i) · Solve by intervals

1. If $x \leq -1$, both distances simplify to $-(x+1)$ and $-(x-1)$; their difference is -2, so there is no solution there.
2. If $-1 \leq x \leq 1$, the expression is $(x+1) - (1-x) = 2x$. Setting $2x = 1$ gives $x = \frac{1}{2}$, which lies in this interval.
3. If $x \geq 1$, the expression is $(x+1) - (x-1) = 2$, not 1.
4. **Answer:** $\{\frac{1}{2}\}$. The interval check is essential: each simplified equation is valid only in its own case.

Exercise 2.1: Absolute values and inequalities

1b(ii) · Domain first, then square

1. The radicand requires $x - 12 \geq 0$, so $x \geq 12$. On this domain, $x - 2 \geq 10$, so both sides are non-negative and squaring preserves the inequality.
2. Square and simplify:

$$x - 12 < (x - 2)^2 \Leftrightarrow 0 < x^2 - 5x + 16$$

3. The quadratic is always positive: its discriminant is $25 - 64 = -39 < 0$ and its leading coefficient is positive. Thus every x in the domain works.
4. **Answer:** $[12, \infty)$. For instance, $x = 12$ gives $0 < 10$, confirming the boundary is included.

Exercise 2.2: Supremum and infimum

2a · The half-open interval

1. Every member is at most 1, and 1 belongs to M_1. Therefore $\sup M_1 = \max M_1 = 1$.
2. Every member is positive, so 0 is a lower bound. Values can be chosen arbitrarily close to 0 from above, so no larger number is a lower bound. Thus $\inf M_1 = 0$.
3. Because 0 is not in the interval, M_1 has no minimum. **Summary:** sup = max = 1; inf = 0; no minimum.

Exercise 2.2: Supremum and infimum

2a · A decreasing sequence of reciprocals

1. Here $\mathbb{N} = \{1, 2, \ldots\}$, as defined in the MEI notes. The first term is 1, and $\frac{1}{n} \leq 1$ for every positive n, so $\sup M_2 = \max M_2 = 1$.
2. All terms are positive. Given any $\varepsilon > 0$, choose a natural $n > \frac{1}{\varepsilon}$; then $\frac{1}{n} < \varepsilon$. Hence no positive number is a lower bound, and $\inf M_2 = 0$.
3. Zero is not one of the terms, so there is no minimum. **Summary:** sup = max = 1; inf = 0; no minimum.

Exercise 2.2: Supremum and infimum

2a · Separate positive and negative terms

1. For odd n, the terms are negative: $-(1 + \frac{1}{n})$. Their smallest value occurs at the smallest odd n, $n = 1$, giving -2. All later negative terms are greater than -2.
2. For even n, terms are positive: $1 + \frac{1}{n}$. Their largest value occurs at the smallest even n, $n = 2$, giving $\frac{3}{2}$. All later positive terms are smaller.
3. **Summary:** $\inf M_3 = \min M_3 = -2$; $\sup M_3 = \max M_3 = \frac{3}{2}$. Both bounds are attained.

Exercise 2.2: Supremum and infimum

2b · The least upper bound is irrational

1. M is non-empty ($1 \in M$), and every $x \in M$ satisfies $x < \sqrt{2}$. Each rational $q > \sqrt{2}$ is therefore an upper bound. There are infinitely many: for example, choose $q = 2 + \frac{1}{n}$ for each positive integer n; all exceed $\sqrt{2}$ and give distinct rationals.
2. Suppose a rational S were $\sup_{\mathbb{Q}} M$. All members of M lie below $\sqrt{2}$, and rationals in M approach it from below because $\mathbb{Q}$ is dense in $\mathbb{R}$. Thus $\sqrt{2}$ is the real least upper bound.
3. By the given ε-characterisation, a supremum S must be an upper bound, so $S \geq \sqrt{2}$. For every $\varepsilon > 0$, there is $x \in M$ with $x > S - \varepsilon$, while every $x \in M$ is below $\sqrt{2}$; hence $S - \varepsilon < \sqrt{2}$ for every ε, forcing $S \leq \sqrt{2}$. Therefore $S = \sqrt{2}$. But that is irrational, contradicting $S \in \mathbb{Q}$.
4. **Conclusion:** M has infinitely many upper bounds in $\mathbb{Q}$ but no supremum in $\mathbb{Q}$. In $\mathbb{R}$ its supremum is $\sqrt{2}$.

Exercise 2.3: Sums, products, binomial coefficients

3a · Index the odd bases

1. The odd integers in the stated range are 11, 13, …, 49. Starting with $k = 0$, the kth base is $11 + 2k$. It reaches 49 when $k = 19$.
2. Therefore the requested sum is:

$$\sum_{k=0}^{19} (11 + 2k)^2$$

3. **Check:** the first term is $11^2 = 121$ and the last is $(11 + 38)^2 = 49^2 = 2401$; there are 20 odd squares.

Exercise 2.3: Sums, products, binomial coefficients

3b(i) · Coefficient cancellation

1. Re-index the first sum with $j = k + 1$. The bracket becomes $\sum_{j=1}^{10} jx^j - \sum_{j=0}^{10} (j+1)x^j$.
2. The constant term is -1. For each power x^j, $1 \le j \le 10$, the coefficient is $j - (j+1) = -1$. Thus the bracket is $-(1 + x + \ldots + x^{10})$.
3. Multiply by $(1 - x)$ and use the finite geometric-sum identity:

$$-(1 - x)(1 + x + \ldots + x^{10}) = -(1 - x^{11}) = x^{11} - 1$$

4. **Answer:** $x^{11} - 1$, valid for every real x, including $x = 1$ because the polynomial identity has no division by $1 - x$.

Exercise 2.3: Sums, products, binomial coefficients

3b(ii) · A zero term removes one product

1. At $m = 1$, the product includes $k = 1$, whose factor is $1^2 - 1 = 0$. That whole inner product is 0.
2. At $m = 2$, the product has $k = 2$ and $k = 3$:

$$(2^2 - 1)(3^2 - 1) = 3 \cdot 8 = 24$$

3. Add the two outer terms: $0 + 24 = \mathbf{24}$.

Exercise 2.3: Sums, products, binomial coefficients

3b(iii) · Count all subsets

1. Expand $(1 + 1)^n$ by the binomial theorem:

$$(1 + 1)^n = \sum_{k=0}^{n} \mathrm{C}(n, k) \cdot 1^{n-k} \cdot 1^k$$

2. Every power of 1 equals 1, so the requested sum equals 2^n. This also counts all subsets of an n-element set.
3. **Answer:** 2^n.

Exercise 2.3: Sums, products, binomial coefficients

3c · Count unordered pairs

1. With g guests, every pair clinks once, so the number of clinks is $\mathrm{C}(g,2) = \dfrac{g(g-1)}{2}$.
2. Set the count to 66 and solve:

$$\frac{g(g-1)}{2} = 66 \Leftrightarrow g^2 - g - 132 = 0 \Leftrightarrow (g-12)(g+11) = 0$$

3. The roots are 12 and -11; only a positive guest count is possible. **There are 12 guests.**
4. **Check:** $12 \cdot \dfrac{11}{2} = 66$ distinct pairs.

Exercise 2.3: Sums, products, binomial coefficients

3d · Use the complement probability

1. Assume independent, equally likely birthdays on 365 days, as specified. The first student can have any birthday; for no match, the next has 364 choices, then 363, continuing down to 339 choices for the 27th student.
2. There are 365^{27} equally likely birthday assignments overall. Hence:

$$\mathrm{P}(\text{all birthdays different}) = \frac{365 \cdot 364 \cdot \ldots \cdot 339}{365^{27}}$$

$$\mathrm{P}(\text{at least one shared birthday}) = 1 - \frac{365 \cdot 364 \cdot \ldots \cdot 339}{365^{27}} \approx 0.6269$$

3. **Answer:** about 62.7%. This is the probability of at least one matching pair, not the expected number of matching pairs.

Exercise 2.4: Order axioms

4i · Distributivity and cancellation

1. Since $0 = 0 + 0$, distributivity gives $a \cdot 0 = a(0+0) = a \cdot 0 + a \cdot 0$.
2. Add the additive inverse of $a \cdot 0$ to both sides. The left becomes 0 and the right becomes $a \cdot 0$, so $\boldsymbol{a \cdot 0 = 0}$.
3. This argument works for every a and uses only the additive identity, distributivity and additive inverses.

Exercise 2.4: Order axioms

4ii · Identify the additive inverse

1. Distributivity gives $(-1)\cdot a + a = ((-1) + 1)\cdot a$.
2. $(-1) + 1 = 0$, and part i) gives $0\cdot a = 0$. Therefore $(-1)\cdot a + a = 0$.
3. An additive inverse is unique, so $\boldsymbol{(-1)\cdot a = -a}$.

Exercise 2.4: Order axioms

4iii · Two negative factors

1. Part ii) gives $-a = (-1)a$ and $-b = (-1)b$. Associativity and commutativity of multiplication then give $(-a)(-b) = ((-1)(-1))(ab)$.
2. Apply part ii) to the element -1: $-(-1) = (-1)(-1)$. The additive inverse of -1 is 1, so this product equals 1.
3. Thus $(-a)(-b) = 1\cdot ab = \boldsymbol{ab}$.

Exercise 2.4: Order axioms

4iv · Use the inverse of a non-zero factor

1. Suppose $ab = 0$. If $a = 0$, the conclusion already holds.
2. Otherwise $a \neq 0$, so a^{-1} exists in the field. Multiply $ab = 0$ by a^{-1}:

$$a^{-1}(ab) = a^{-1} \cdot 0$$

3. Associativity and the zero-product result from i) reduce this to $b = 0$. In either case, $\boldsymbol{a = 0}$ **or** $\boldsymbol{b = 0}$.
4. The existence of inverses for non-zero elements is the crucial field property; it need not hold in an arbitrary ring.

Exercise 2.4: Order axioms

4v · Exclude the other order cases

1. In a field, $1 \neq 0$. Trichotomy therefore leaves either $1 > 0$ or $1 < 0$.
2. Assume for contradiction that $1 < 0$. Add -1 to both sides: $0 < -1$.
3. The product of two positive elements is positive, so $(-1)(-1) > 0$. Part iii), or the identity established in its proof, says $(-1)(-1) = 1$. Thus $1 > 0$, contradicting $1 < 0$.
4. The negative case is impossible. Therefore $\boldsymbol{1 > 0}$.

Exercise 2.4: Order axioms

4vi · A non-zero square is positive

1. By trichotomy, $a \neq 0$ means either $a > 0$ or $a < 0$.
2. If $a > 0$, the product of positive elements gives $a \cdot a > 0$.
3. If $a < 0$, then $-a > 0$. Its square is positive, and part iii) gives $(-a)(-a) = a \cdot a$.
4. Both cases establish $\boldsymbol{a^2 > 0}$.

Exercise 2.4: Order axioms

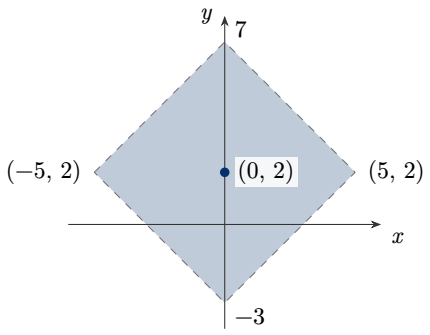
4vii · Multiplication by a negative reverses order

1. Add $-a$ to the inequality $a < b$. Compatibility of order with addition gives $0 < b - a$.
2. Because $-c > 0$ and $b - a > 0$, the product $(-c)(b - a)$ is positive. The identity $(-c)(b - a) = -(c(b - a))$ follows from distributivity and additive inverses, so $c(b - a) < 0$.
3. Expand: $c(b - a) = cb - ca < 0$. Add ca to both sides to obtain $cb < ca$.
4. Commutativity gives $\boldsymbol{ac > bc}$. The direction reverses because c is negative.

Self-Study 2.5: Absolute values and inequalities

5a · Interior of a translated diamond

1. $|x|$ is horizontal distance from $x = 0$; $|y - 2|$ is vertical distance from $y = 2$. Their sum is the Manhattan distance from $(0, 2)$.
2. Set one distance to zero to find the vertices: $x = \pm 5$ when $y = 2$; and $y - 2 = \pm 5$ when $x = 0$, giving $y = 7$ and $y = -3$.
3. The strict inequality keeps the interior of the diamond with vertices $(-5,2)$, $(5,2)$, $(0,7)$, and $(0,-3)$. The boundary is excluded.



$\boldsymbol{|x| + |y - 2| < 5}$. The shaded interior, centred at $(0, 2)$, is the solution set; the dashed boundary is excluded.

Self-Study 2.5: Absolute values and inequalities

5b(i) · Sum of distances from two points

1. For $-3 \leq x \leq 3$, $|x{+}3| = x{+}3$ and $|x{-}3| = 3{-}x$. Their sum is 6, which is not greater than 8.
2. For $x > 3$, both terms are positive expressions, so the sum is $(x{+}3){+}(x{-}3){=}2x$. The inequality gives $x > 4$.
3. For $x < -3$, both absolute values reverse sign and the sum is $-2x$. Then $-2x > 8$ gives $x < -4$.
4. **Answer:** $(-\infty, -4) \cup (4, \infty)$. The endpoints are excluded because the original inequality is strict.

Self-Study 2.5: Absolute values and inequalities

5b(ii) · Combine domain restrictions, then square

1. The radicand is non-negative when $x \leq 1$ or $x \geq 4$. Since a square root is non-negative, the right side $2 - x$ must also be non-negative, so $x \leq 2$. Together these restrictions reduce the domain to $x \leq 1$.
2. On this domain both sides are non-negative, so squaring is equivalent:

$$(x - 4)(x - 1) \leq (2 - x)^2 \Leftrightarrow x^2 - 5x + 4 \leq x^2 - 4x + 4 \Leftrightarrow x \geq 0$$

3. Intersect with $x \leq 1$ to obtain $\mathbf{[0,1]}$.
4. **Check:** at 0, $\sqrt{4} = 2$, equality holds; at 1, $0 \leq 1$. Values below 0 fail the squared inequality.

Self-Study 2.6: Sets, products and complements

6a · Element-chasing proof

1. Let $(a,b) \in A \times B$. By definition, $a \in A$ and $b \in B$.
2. $A \subset C$ gives $a \in C$, and $B \subset D$ gives $b \in D$.
3. Therefore $(a,b) \in C \times D$. Since the chosen pair was arbitrary, $\boldsymbol{A \times B \subset C \times D}$.

Self-Study 2.6: Sets, products and complements

6b · Empty products hide factor membership

1. The converse would claim $A \times B \subset C \times D$ implies both $A \subset C$ and $B \subset D$. This is false when a factor is empty.
2. For example, let $A = \{1\}$, $B = \varnothing$, $C = \varnothing$, and $D = \{2\}$. Then $A \times B = \varnothing$ and $C \times D = \varnothing$, so $A \times B \subset C \times D$, but A is not a subset of C.
3. If A and B are both non-empty, choose $b \in B$ to test any $a \in A$, and choose $a \in A$ to test any $b \in B$. The product inclusion then does force $A \subset C$ and $B \subset D$.
4. **Conclusion:** not in general; it becomes true if both factors A and B are non-empty.

Self-Study 2.6: Sets, products and complements

6c · Track open and closed endpoints

1. A contains 1 but not 3, so its complement contains values below 1 and values at least 3:

$$A^c = (-\infty, 1) \cup [3, \infty)$$

2. B excludes 2 and includes 5:

$$B^c = (-\infty, 2] \cup (5, \infty)$$

3. The overlap intervals join into $A \cup B = [1,5]$. Therefore $(A \cup B)^c = (-\infty,1) \cup (5,\infty)$.
4. The intersection is $(2,3)$, because both endpoints are excluded by at least one interval. Thus $(A \cap B)^c = (-\infty,2] \cup [3,\infty)$.
5. $A \setminus B$ contains the part of A not in B: $[1,2]$. The point 2 remains because B is open there; every point just above 2 is removed.
6. **Check:** 1 and 2 belong to $A \setminus B$; 3 does not belong to A at all.

Self-Study 2.6: Sets, products and complements

6d · Outside in either coordinate

1. A point (x,y) is in the rectangle exactly when $1 \leq x \leq 3$ and $1 \leq y \leq 2$.
2. Negate this conjunction: $x < 1$ or $x > 3$ or $y < 1$ or $y > 2$.
3. Thus the complement is:

$$((-\infty, 1) \cup (3, \infty)) \times \mathbb{R} \cup \mathbb{R} \times ((-\infty, 1) \cup (2, \infty))$$

4. The rectangle edges are not in the complement because the original intervals are closed.

Self-Study 2.7: Sums, products, binomial coefficients

7a · Encode both sequences with one index

1. For $k = 0,1,2,3,4$, the positive base $10 - 2k$ runs 10,8,6,4,2; the denominator $9 - 2k$ runs 9,7,5,3,1.
2. Therefore the sum is:

$$\sum_{k=0}^{4} \left((10 - 2k)^2 - \frac{1}{(9 - 2k)^2} \right)$$

3. **Check:** the final term is $2^2 - \frac{1}{1^2}$, as required.

Self-Study 2.7: Sums, products, binomial coefficients

7b(i) · Shifted finite sums

1. The first sum is $1 + q + \ldots + q^n$. The second is $q^{-1} + 1 + q + \ldots + q^n$.
2. Every term cancels except the initial q^{-1} in the second sum, which is subtracted:

$$\sum_{k=0}^{n} q^k - \sum_{k=0}^{n+1} q^{k-1} = -q^{-1} = -\frac{1}{q}$$

3. This holds for every $q \neq 0$, including $q = 1$; no geometric-series division by $q - 1$ is needed.

Self-Study 2.7: Sums, products, binomial coefficients

7b(ii) · Two telescoping products

1. Factor each term:

$$1 - \frac{2}{j(j+1)} = \frac{j^2 + j - 2}{j(j+1)} = \frac{(j-1)(j+2)}{j(j+1)}$$

2. Split the product into $\prod_{j=2}^{n} \frac{j-1}{j}$ and $\prod_{j=2}^{n} \frac{j+2}{j+1}$. The first collapses to $\frac{1}{n}$; the second to $\frac{n+2}{3}$.
3. Multiply:

$$\prod_{j=2}^{n} \left(1 - \frac{2}{j(j+1)} \right) = \frac{n+2}{3n}$$

4. The formula also gives 1 at $n = 1$, agreeing with the empty-product convention.

Self-Study 2.7: Sums, products, binomial coefficients

7b(iii) · Binomial theorem at $x = -1$

1. The binomial theorem says $(1 + x)^n = \sum_{k=0}^{n} \mathrm{C}(n,k) x^k$.
2. Set $x = -1$. For $n \geq 1$, the left side is 0, so:

$$\sum_{k=0}^{n} (-1)^k \mathrm{C}(n,\, k) \;=\; (1 - 1)^n \;=\; 0$$

3. **Answer:** 0.

Self-Study 2.7: Sums, products, binomial coefficients

7c · Count the allowed composition

1. At most two striped means 0, 1, or 2 striped marbles. But six marbles cannot be chosen with fewer than two striped because the box has only four white marbles.
2. Therefore exactly two striped and all four white must be selected. Choose 2 of the 12 striped marbles and 4 of the 4 white ones:

$$\mathrm{C}(12,\, 2) \cdot \mathrm{C}(4,\, 4) \;=\; 66 \cdot 1 \;=\; 66$$

3. **Answer:** 66 ways.

Self-Study 2.7: Sums, products, binomial coefficients

7d · Count ordered speaker-listener pairs

1. Choose the speaker in m ways. That member then greets each of the other $m - 1$ members once, so the total number of spoken riddles is $m(m - 1)$, not $\mathrm{C}(m,2)$. Speaking and listening are different roles, so the two directions between a pair count separately.
2. Set the ordered-pair count to 272 and solve:

$$m(m - 1) \;=\; 272 \;\Leftrightarrow\; m^2 - m - 272 \;=\; 0 \;\Leftrightarrow\; (m - 17)(m + 16) \;=\; 0$$

3. The roots are 17 and -16; only a positive group size is possible. **There are 17 members.**
4. **Check:** each of 17 members speaks to 16 others, giving $17 \cdot 16 = 272$ spoken riddles. If the greetings were mutual and each pair spoke only once, the count would instead be $\mathrm{C}(m,2)$, as in Exercise 2.3c.

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